AP PRECALCULUS

Unit 1 • Lesson 1.2 • Guided notes

Topic 1.2 • Rates of Change

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Compute the average rate of change of a function over an interval, from a formula, a table or a graph.
- Interpret an average rate of change in context, with units.
- Estimate the rate of change at a point, and relate its sign to increasing and decreasing behavior.

Vocabulary. average rate of change, secant line, slope, rate of change at a point, units.

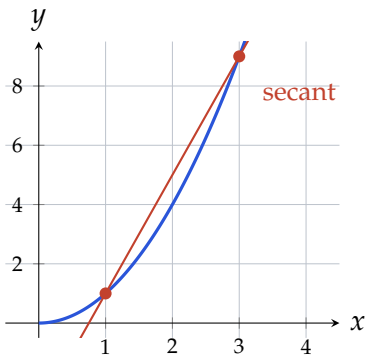
1. Average rate of change

Key idea

The average rate of change of f over the interval $[a, b]$ is

$$\frac{\text{_____}}{\text{_____}}.$$

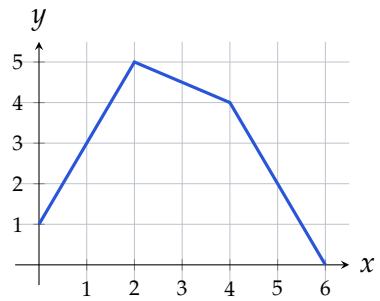
It is the slope of the _____ line through the points $(a, f(a))$ and $(b, f(b))$. Its units are _____ per _____.



Example 1 • From a formula

Find the average rate of change of $f(x) = x^2$ on $[1, 3]$, and of $g(x) = x^3 - 4x$ on $[-2, 1]$.

Example 2 • From a graph



Find the average rate of change of h on $[0, 2]$, $[2, 6]$ and $[0, 6]$.

2. Interpreting in context

Template
"On average, [quantity] _____ by [number] [output units] per [input unit] from [input a] to [input b]."

Example 3 • Units

$V(t)$ is the number of visitors to a website t hours after midnight; $V(2) = 150$ and $V(6) = 870$. Compute and interpret $\frac{V(6)-V(2)}{6-2}$.

3. Rate of change at a point

The rate of change at $x = a$ can be _____ by average rates of change over _____ intervals that contain a, using values on both sides.

Example 4 • Estimating

Estimate the rate of change of $f(x) = x^3$ at $x = 2$.

Sign of the rate of change
If the rate of change is positive on an interval, f is _____ there; if it is negative, f is

_____. At a point where f changes from increasing to decreasing, the rate of change changes from _____.

Common mistakes

- Subtracting in different orders: $\frac{f(b)-f(a)}{a-b}$.
- Giving the change $f(b) - f(a)$ instead of the rate.
- Forgetting units in an interpretation.

Check your understanding

1. The average rate of change of $g(x) = 5 - x^2$ on $[-1, 2]$ is _____.
2. If $T(4) = 21$ and $T(10) = 9$ (°C, hours), the average rate of change on $[4, 10]$ is _____.